ScienceX

Battery Storage Market Notes, October 2026 issue

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The price of electricity changes with the time of day. "Grid-scale batteries" earn money from that difference. These are large batteries connected to the grid that store electricity and release it later. These notes use published figures and our own records to show this month's trends in figures and tables.

1. The three income sources of a battery storage site (how to read these notes)

A battery storage site (a grid-scale battery) earns income from three markets. ① Wholesale electricity (JEPX) is the market where the site buys electricity at cheap times, stores it, and sells it at expensive times. ② The balancing market (EPRX) is the market where the site sells grid operators the "capability" to correct surpluses and shortages of electricity. ③ The capacity market (OCCTO) is the market where the government buys future electricity supply capacity in advance. These notes present figures for these three markets in this order. The ①②③ at the start of each section title shows which market the section covers.

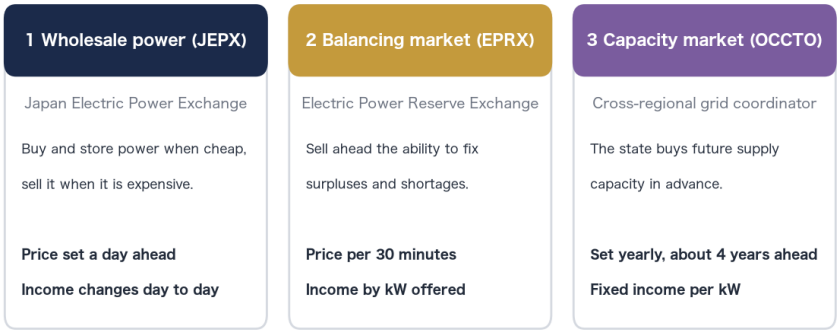


Figure 1 The three income sources of a battery storage site

Source: The market rules of the Japan Electric Power Exchange (JEPX), the Electric Power Reserve Exchange (EPRX) and the Organization for Cross-regional Coordination of Transmission Operators (OCCTO).

2. Three trends this month

We picked three trends this month from the figures of the three markets.

- In wholesale electricity (JEPX), the spread between the average of the 4 most expensive hours and the average of the 4 cheapest hours (monthly average) in October 2026 was 17.14 yen/kWh in Tokyo and 19.30 yen/kWh in Kyushu.
- In the balancing market (EPRX), the bid ratio for Primary reserve (online) went above 100% in June 2026, at 106.8%. In September 2026 it was 123.1%, and the average awarded price was 3.43 yen/kW per 30 min.
- In the balancing market (EPRX), the top 10% of 30-minute slots by sales accounted for 56.9% of the month's sales for Tertiary reserve 2 in September 2026.

3. ① Wholesale electricity (JEPX) | Average price

Wholesale electricity is the electricity that power companies and retailers buy and sell among themselves. Its price is set on the "spot market" of the Japan Electric Power Exchange (JEPX), where electricity for the next day is traded the day before. "System (nationwide)" is the price when all of Japan is treated as one market (the system price). When the power lines between regions are congested, prices split by area, from Hokkaido to Kyushu (the area prices).

The monthly average of the System (nationwide) price in wholesale electricity (JEPX) was 22.51 yen/kWh in October 2026 and 11.67 yen/kWh in the same month last year, a difference of 10.84 yen.



Figure 2 Average wholesale electricity price (yen/kWh). Navy is October 2026, grey is the same month last year

Source: Spot market trading results published by the Japan Electric Power Exchange (JEPX). Simple average of the price for each 30-minute slot. Our own aggregation.

4. ① Wholesale electricity (JEPX) | How the price moves during the day

The wholesale electricity price changes every 30 minutes. The figure shows the shape of a day, made by averaging the prices at each time of day across the month. If a battery charges at cheap times and discharges at expensive times, the difference becomes its income.

In October 2026, the average wholesale price during the day ranged from a low of 19.03 yen around 12:00 to a high of 37.11 yen around 16:30 in Tokyo. In Kyushu it ranged from 4.94 yen around 12:00 to 24.74 yen around 17:30.

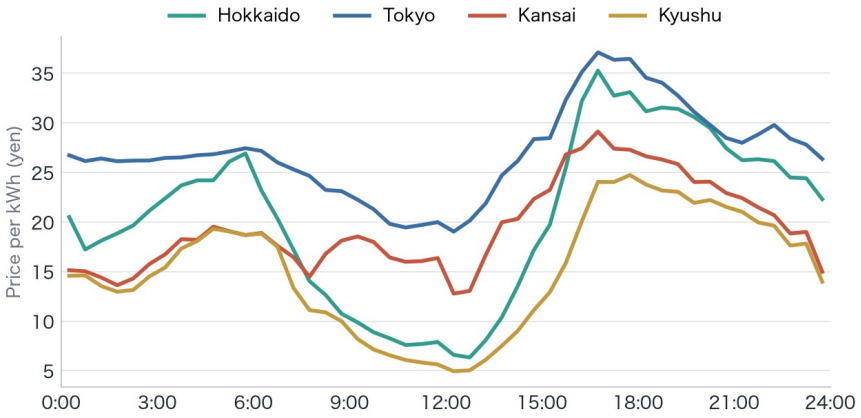


Figure 3 Peaks and valleys of the daily wholesale electricity price (October 2026 average, yen/kWh)

Source: JEPX spot market. Our own aggregation.

5. ① Wholesale electricity (JEPX) | How large the spread is per kWh

For each day, we took the average price of the 4 most expensive hours and subtracted the average price of the 4 cheapest hours. We show the monthly average by area. This equals the difference per kWh if you buy 1 kWh in the 4 cheapest hours and sell it in the 4 most expensive hours. It is a simple difference. It does not subtract electricity lost in charging and discharging or fees, and it ignores whether the cheap hours come before the expensive hours.

The October 2026 spread between the 4 most expensive and 4 cheapest hours in wholesale electricity (monthly average) was largest in Hokkaido, at 29.02 yen/kWh, and smallest in Tokyo, at 17.14 yen/kWh.

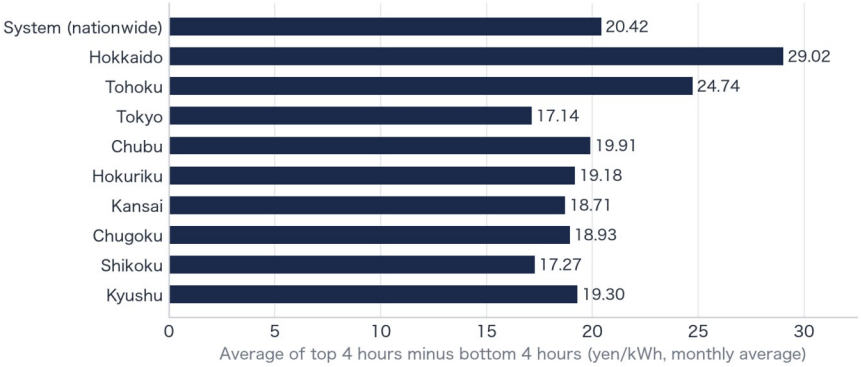


Figure 4 Wholesale electricity spread: average of the 4 most expensive hours − average of the 4 cheapest hours (October 2026, yen/kWh, monthly average)

Source: JEPX spot market. Our own aggregation.

Area	2-hour spread (yen/kWh)	4-hour spread (yen/kWh)	Days with a 4-hour spread of 10 yen or more
System (nationwide)	22.95	20.42	29
Hokkaido	32.56	29.02	31
Tohoku	27.09	24.74	30
Tokyo	19.62	17.14	29
Chubu	22.37	19.91	29
Hokuriku	21.71	19.18	31
Kansai	21.22	18.71	31
Chugoku	20.90	18.93	31
Shikoku	19.40	17.27	28

Area	2-hour spread (yen/kWh)	4-hour spread (yen/kWh)	Days with a 4-hour spread of 10 yen or more
Kyushu	21.19	19.30	31

The spread is the monthly average of the daily spreads. October 2026 covers 31 days.

6. ① Wholesale electricity (JEPX) | There is a spread every day

We show the same 4-hour spread day by day. The larger the spread on a day, the more room a battery has to earn from it.

In October 2026, out of 31 days, the 4-hour wholesale spread was 10 yen or more on 29 days for System (nationwide), on 31 days in Hokkaido, Hokuriku, Kansai, Chugoku and Kyushu, and on 28 days in Shikoku.

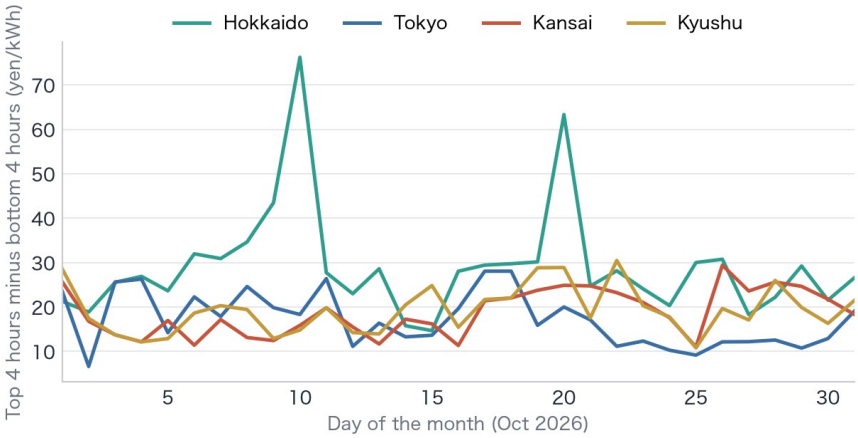


Figure 5 Daily wholesale electricity spread (average of the day's 4 most expensive hours − average of the 4 cheapest hours, yen/kWh)

Source: JEPX spot market. Our own aggregation.

7. ② Balancing market (EPRX) | Seven products, and prices by area

The balancing market is where grid operators buy, every 30 minutes, the "capability" to correct surpluses and shortages of electricity. Products differ by how fast output must change after a dispatch instruction. Primary reserve is the fastest, and Tertiary reserve 2 is the slowest. Primary reserve has two categories. "Online" is connected to the grid operator by a dedicated line. "Offline" runs on the simplified dispatch system without a dedicated line. The two are procured and traded separately, so we show them separately. A Composite product is a bid that offers several products together. Prices are set in yen per 1 kW of capability per 30 minutes. The figure shows the range of awarded prices by area (by the area where the power source is located). The thin line runs from the lowest to the highest awarded bid in the month. The thick band is the price range that contains 8 out of every 10 slots with awards. The dot is the average awarded price. Nationwide figures are in the table in the "Market figures" section. Published figures for October 2026 are not yet in our aggregation, so we show the latest month with all products available: September 2026 (1–29), preliminary figures.

In September 2026, the average awarded price in the balancing market, by the area where the power source is located, was highest for Primary reserve (offline) in Tohoku, at 9.51 yen/kW per 30 min, and lowest for Tertiary reserve 2 in Shikoku, at 0.38 yen/kW per 30 min.

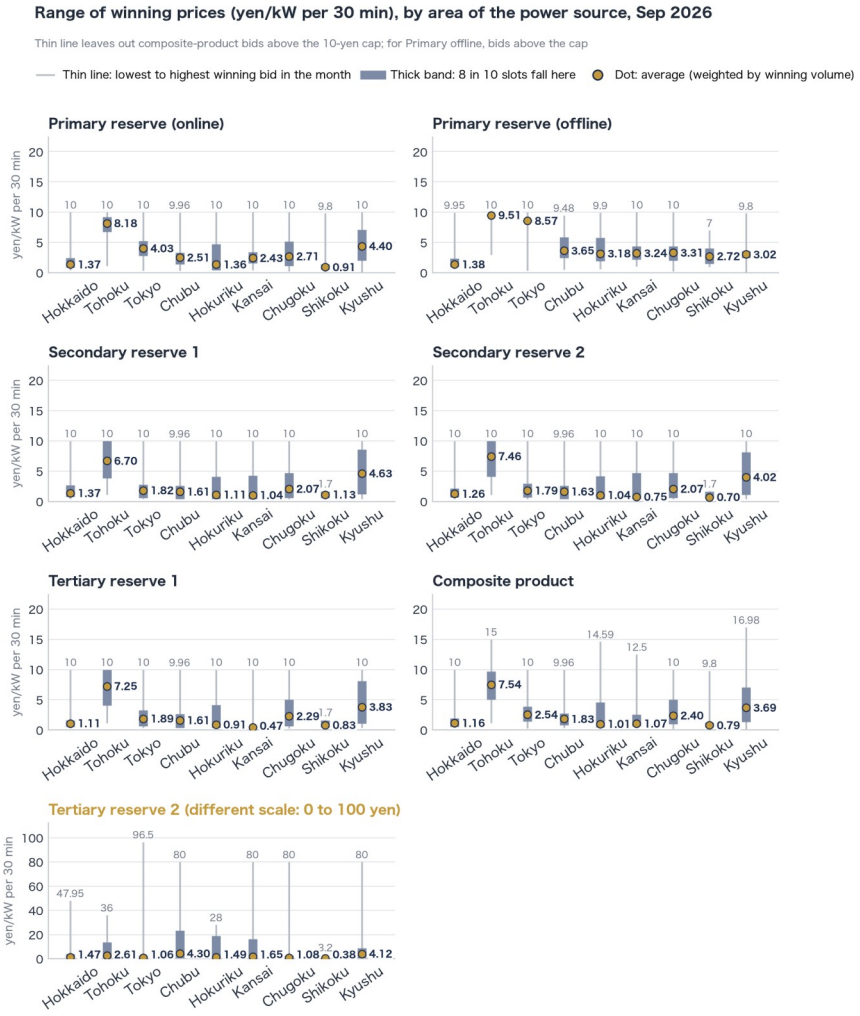


Figure 6 Range and average of awarded prices by area in the balancing market (September 2026 (1–29), by the area where the power source is located)

Source: Trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our own aggregation.

Product	Time from instruction to output change	Duration
Primary reserve (online)	Within 10 seconds	5 minutes or more
Primary reserve (offline)	Within 10 seconds	5 minutes or more
Secondary reserve 1	Within 5 minutes	30 minutes
Secondary reserve 2	Within 5 minutes	30 minutes
Tertiary reserve 1	Within 15 minutes	30 minutes
Tertiary reserve 2	Within 60 minutes	30 minutes
Composite product	Requirements of the combined products	Requirements of the combined products

8. ② Balancing market (EPRX) | Grid areas and times where bids are accepted more easily

The clearing rate is the share of the bid volume that was awarded. The higher it is, the more easily bids are accepted. The figure shows the clearing rate for each product by time of day (every 30 minutes) in each grid area (the area where the power source is located). Where a line breaks, there were no bids at that time.

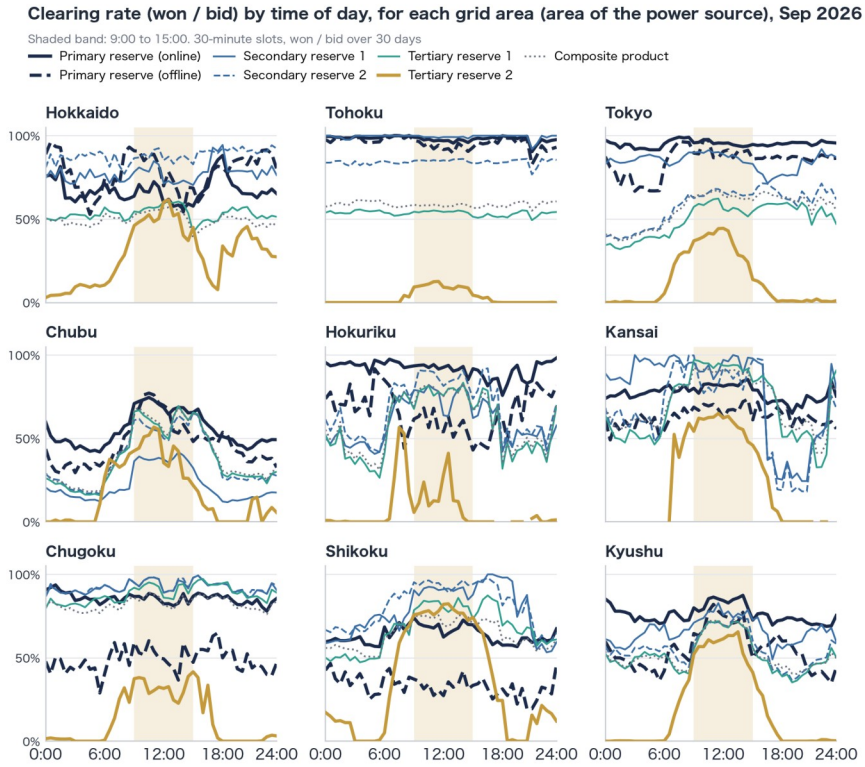


Figure 7 Clearing rate in the balancing market (grid area (area where the power source is located) × time of day, September 2026 (1–29))

Source: Trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our own aggregation.

9. ② Balancing market (EPRX) | Primary reserve (online) became a competition for seats

Primary reserve is the fastest product. Output must change within 10 seconds of an instruction, and batteries are often used for it. The left side is the online category, and the right side is the offline category. The bars show the bid ratio (bid volume ÷ volume procured by grid operators). Above 100%, more bids come in than are procured, and some bids are rejected. The bars rise when bids increase and also when the procured volume falls. The line shows the average awarded price. These are monthly figures from April 2025, the data we have.

The bid ratio for Primary reserve (online) in the balancing market went above 100% in June 2026, at 106.8%, and was 123.1% in September 2026. The average awarded price went from 2.89 yen/kW per 30 min in April 2025 to 3.43 yen/kW per 30 min in September 2026. The procured volume went from 3,066.1MW in April 2025 to 1,388.7MW in September 2026. For Primary reserve (offline), the bid ratio in September 2026 was 84.0% and the average awarded price was 5.79 yen/kW per 30 min. The procured volume went from 1,122.5MW in April 2025 to 497.8MW in September 2026.

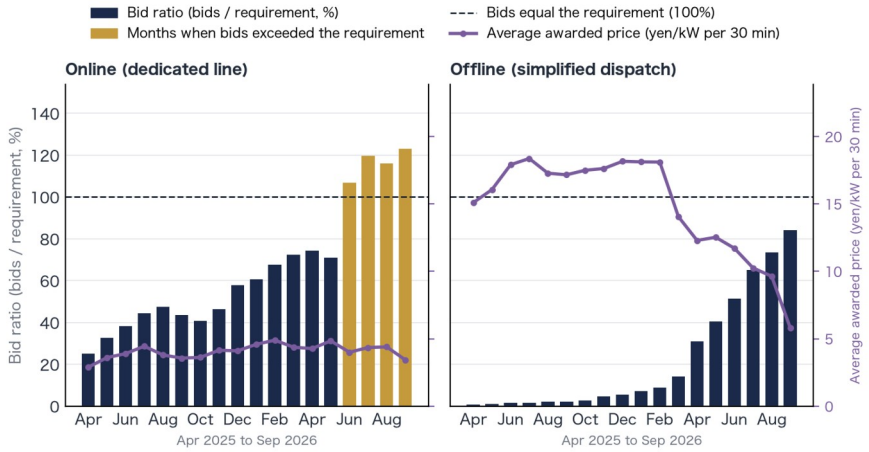


Figure 8 Primary reserve bid ratio (bars) and average awarded price (line) (monthly; left is online, right is offline; gold bars are months when bids exceeded the procured volume)

Source: Trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our own aggregation.

10. ② Balancing market (EPRX) | How prices by product have moved

These are average awarded prices by product from April 2025 (monthly averages weighted by awarded volume). From March 14, 2026, Primary reserve through Tertiary reserve 1 and the Composite product changed from trading every 3 hours to trading every 30 minutes (Tertiary reserve 2 was already traded every 30 minutes).

From April 2025 to September 2026, the average awarded price in the balancing market (yen/kW per 30 min) went from 2.89 yen to 3.43 yen for Primary reserve (online), from 15.10 yen to 5.79 yen for Primary reserve (offline), and from 1.12 yen to 2.22 yen for Tertiary reserve 2. The highest month was February 2026 for Primary reserve (online), at 4.90 yen, July 2025 for Primary reserve (offline), at 18.37 yen, and May 2026 for Tertiary reserve 2, at 4.18 yen.

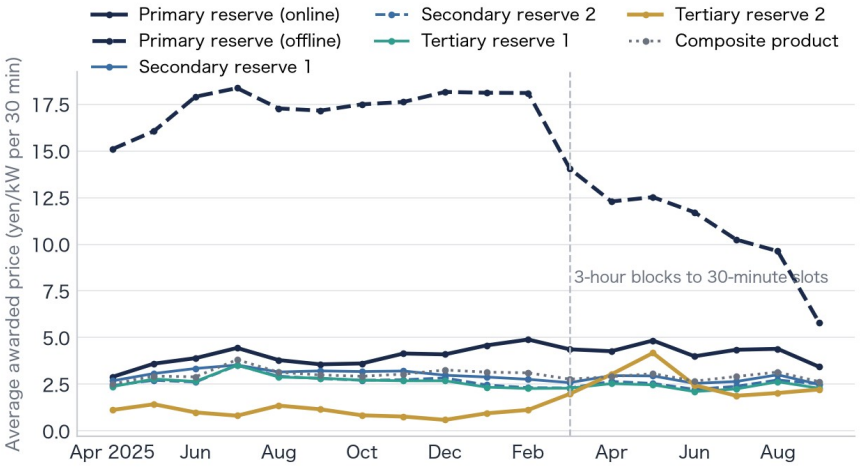


Figure 9 Average awarded price by product in the balancing market (yen/kW per 30 min, monthly)

Source: Trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our own aggregation.

11. ② Balancing market (EPRX) | Which hours of the day are expensive

These are the average awarded prices by product for each hour in September 2026 (1–29).

In September 2026, the average awarded price by hour in the balancing market was highest around 18:00, at 4.09 yen, and lowest around 0:00, at 2.75 yen, for Primary reserve (online). For Tertiary reserve 2 it was highest around 19:00, at 5.02 yen, and lowest around 0:00, at 0.47 yen (yen/kW per 30 min).

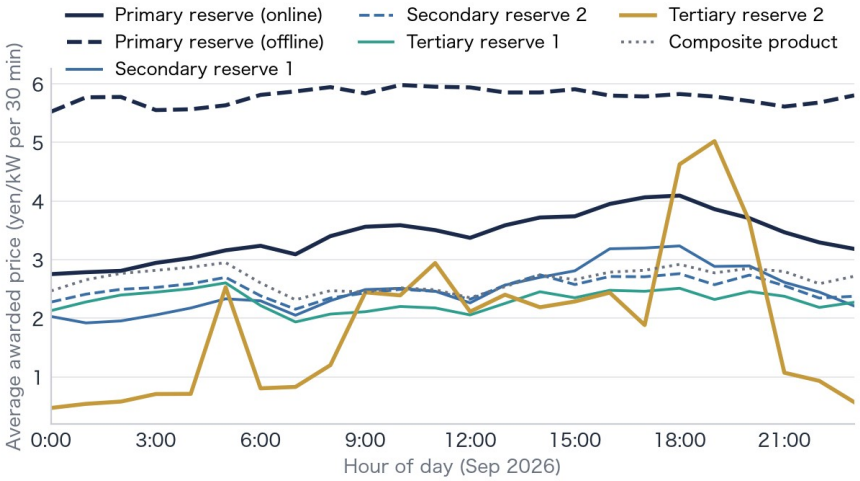


Figure 10 Average awarded price by hour in the balancing market (September 2026 (1–29), yen/kW per 30 min)

Source: Trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our own aggregation.

12. ② Balancing market (EPRX) | Which products batteries are used for

The figure shows the share of the awarded volume taken by batteries. The table compares the average awarded price of the volume awarded to batteries with the average awarded price of the whole product.

In September 2026, the battery share in the balancing market was 27.9% for Primary reserve (online) and 100.0% for Primary reserve (offline). The average awarded price of the volume awarded to batteries was 5.17 yen/kW per 30 min for Primary reserve (online) (overall 3.43 yen/kW per 30 min) and 59.26 yen/kW per 30 min for Tertiary reserve 2 (overall 2.22 yen/kW per 30 min).

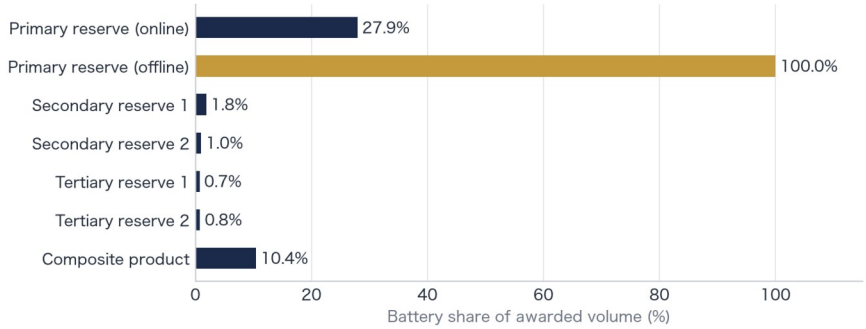


Figure 11 Share of batteries in the balancing market (%)

Source: Trading results (preliminary) by power source type published by the Electric Power Reserve Exchange (EPRX). Our own aggregation.

Product	Volume awarded to batteries (yen/kW per 30 min)	Whole product (yen/kW per 30 min)
Primary reserve (online)	5.17	3.43
Primary reserve (offline)	5.79	5.79
Secondary reserve 1	2.18	2.49
Secondary reserve 2	1.85	2.51
Tertiary reserve 1	1.86	2.28
Tertiary reserve 2	59.26	2.22
Composite product	5.13	2.64

Both are averages weighted by awarded volume (September 2026, preliminary, nationwide). For Tertiary reserve 2, batteries took only 0.8% of the awarded volume, and most of it was awarded in high-price slots, so their average comes out high.

13. ② Balancing market (EPRX) | Sales concentrate in a small number of slots

We ranked the month's 30-minute slots with awards by sales (average awarded price × awarded volume, nationwide), from highest to lowest. For each product, we plotted what % of the month's sales the top slots account for. The more the curve bulges above the dotted line, the more sales concentrate in a few slots. Tertiary reserve 2 is the product that covers forecast errors in renewable output caused by the weather. Its price is low in most slots and high in a very small number of slots.

In September 2026, the top 1% and top 10% of slots by sales in the balancing market accounted for 10.1% and 56.9% of the month's sales for Tertiary reserve 2, and 2.1% and 15.8% for Primary reserve (online).

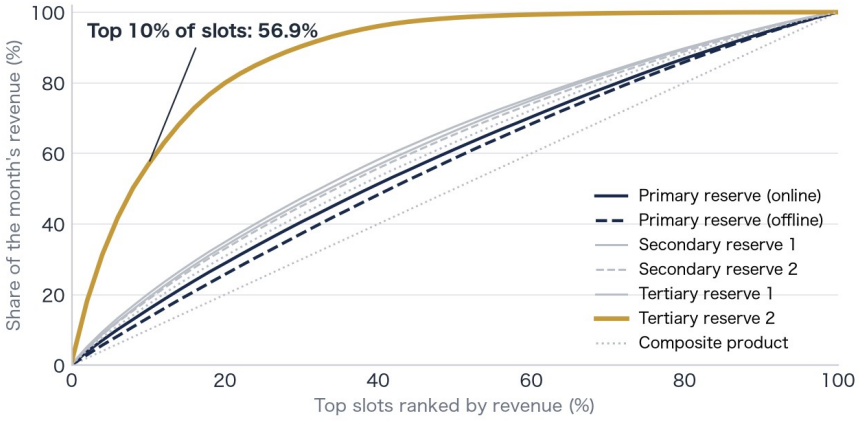


Figure 12 Share of balancing market sales concentrated in the top slots (September 2026 (1–29), nationwide)

Source: Trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our own aggregation.

In September 2026, out of 1,194 slots with awards, the top 1% of slots by sales (average awarded price × awarded volume) accounted for 9.4% of the month's sales. 25 slots had a nationwide average awarded price of 10 yen/kW per 30 min or more, and 198 slots had no awards.

Slots with high maximum awarded prices (by the area where the power source is located):

Date	Time	Area	Maximum awarded price (yen/kW per 30 min)
September 8	9:30	Tokyo	96.50
September 8	10:30	Tokyo	96.50
September 8	11:00	Tokyo	96.50
September 27	11:00	Tokyo	96.00
September 27	11:30	Tokyo	96.00

Source: Tertiary reserve 2 trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our own aggregation.

14. ③ Capacity market (OCCTO) | The market that buys future supply capacity

The capacity market buys electricity supply capacity for 4 years ahead through an auction held once a year. Prices are set in yen per kW per year. The left figure shows the clearing prices of past main auctions. The right figure shows the battery and pumped-hydro capacity contracted in the Long-Term Decarbonization Power Source Auction (a separate auction that supports new builds such as batteries and pumped hydro). Results are published once a year, so they do not change every month. The latest is for delivery FY2029 (published January 2026).

The capacity market main auction clearing price for Tokyo was 14,137 yen/kW for delivery FY2024, 3,495 yen/kW for FY2025, 5,834 yen/kW for FY2026, 9,555 yen/kW for FY2027, 14,812 yen/kW for FY2028 and 15,111 yen/kW for FY2029. The battery and pumped-hydro capacity contracted in the Long-Term Decarbonization Power Source Auction was 1,669 MW in the FY2023 auction; 961 MW for 3–6 hours and 769 MW for 6 hours or more in the FY2024 auction; and 819 MW for lithium-ion and 886 MW for non-lithium in the FY2025 auction. Both are annual values, so they do not change every month; these are the latest published figures.

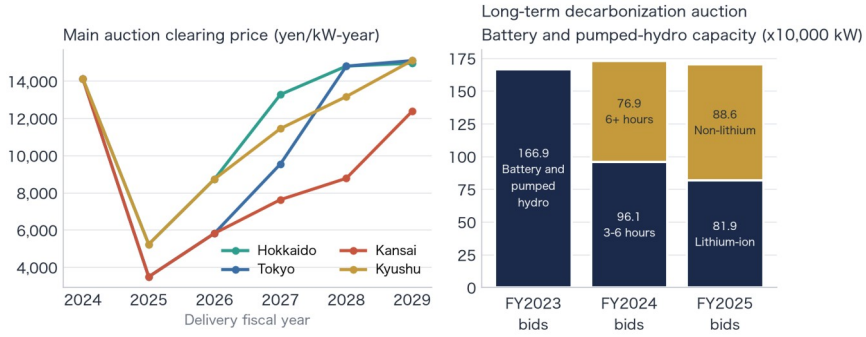


Figure 13 Capacity market clearing prices (left, yen/kW) and battery and pumped-hydro capacity contracted in the Long-Term Decarbonization Power Source Auction (right, in units of 10,000 kW)

Source: Clearing results and capacity contract documents published by the Organization for Cross-regional Coordination of Transmission Operators (OCCTO) (sources are listed in the last section (Assumptions and terms)). As published.

15. Our records | Where battery storage projects are increasing

We counted the projects newly listed this month on our DealRoom (our data room for buyers, a site where we list battery storage projects), by grid area of the grid operator. These are figures for projects we handled, not counts for the whole market. Transfer of rights is a sale that hands over the land and grid connection rights. Completed-build handover is a sale that hands over the site once the equipment is built.

New listings on our DealRoom in October 2026 were 77 projects, 309.1 MW. By grid area, these included Tokyo with 33 projects, 99.8 MW; Tohoku with 8 projects, 73.4 MW; Chubu with 11 projects, 69.9 MW; and Kyushu with 13 projects, 19.9 MW.

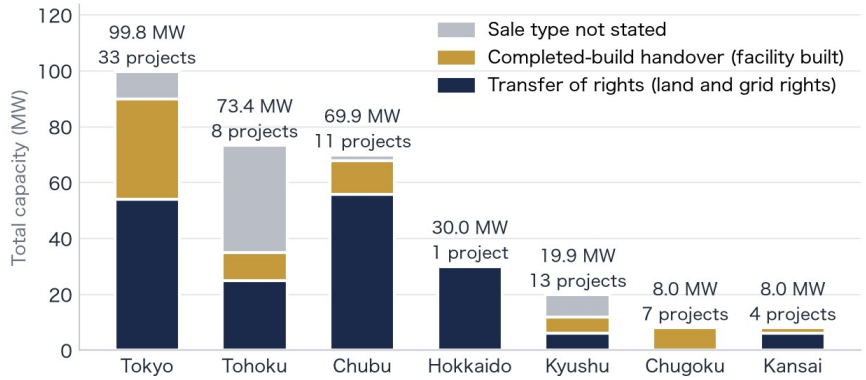


Figure 14 Output of projects listed on our DealRoom in October 2026 (MW)

Source: Our DealRoom records (project locations are shown only down to the grid area; prices and parties are not shown).

16. Our records | How many months and how much it takes to connect to the grid

A grid connection study is the procedure where the grid operator checks "whether this site can connect, and how much the work will cost." The grid connection cost contribution is the part of the connection work cost paid by the applicant. Of the answer documents we have received, we aggregated only high-voltage cases (connected to 6,600V distribution lines), by grid area. Extra-high-voltage cases are not included. Grid areas with fewer than 5 cases are not shown.

The medians of grid connection study answers (high voltage only, as of October 19, 2026) ranged from 8.0 months in Tokyo to 17.0 months in Tohoku for the construction period, and from ¥3.02 million in Tokyo to ¥32.35 million in Tohoku for the grid connection cost contribution (Tohoku: n=5 and n=8).

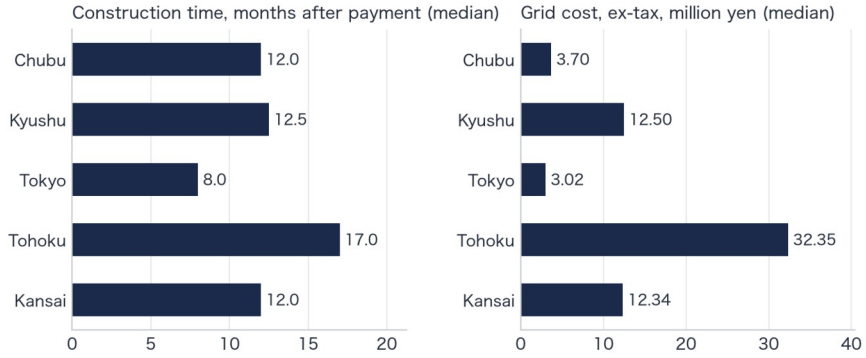


Figure 15 Medians of grid connection study answers (high voltage) by grid area (left: months for the work; right: grid connection cost contribution, excluding tax)

Source: Aggregation of the high-voltage portion of grid connection study answer documents from general transmission and distribution operators that we received (as of October 19, 2026).

17. Market figures (tables you can use as they are)

We put the numbers behind the figures in the text into tables you can use in your own documents. They are ① wholesale electricity (JEPX) figures and trading volume by area, ② nationwide balancing market figures by product (bid ratio, clearing rate, price, battery share), and ③ capacity market clearing prices.

Wholesale electricity, JEPX spot market (October 2026, by area, yen/kWh)

Area	Monthly average	Same month last year	Daytime (8:00–16:00)	Evening (16:00–20:00)	4-hour spread	Highest	Slots at 30 yen or more	Slots at 0.01 yen
System (nationwide)	22.51	11.67	16.02	31.06	20.42	47.88	220	39
Hokkaido	20.49	12.89	11.31	32.25	29.02	105.00	303	67
Tohoku	23.99	12.08	15.78	34.19	24.74	50.01	437	95
Tokyo	27.02	13.06	23.12	34.69	17.14	50.01	466	16
Chubu	25.46	11.68	20.38	33.55	19.91	46.45	387	24
Hokuriku	19.70	11.27	18.29	26.92	19.18	46.45	218	24
Kansai	19.50	11.27	18.22	26.76	18.71	46.45	215	29
Chugoku	15.63	11.12	9.86	23.10	18.93	44.79	63	79
Shikoku	11.44	9.28	7.50	16.61	17.27	37.51	23	146
Kyushu	15.07	10.74	8.31	23.10	19.30	44.79	53	91

Trading volume (October 2026, total of 31 days): total contracted volume 30,319 million kWh (same month last year 22,279 million kWh), sell bid volume 45,218 million kWh, buy bid volume 38,191 million kWh.

Source: Spot market trading results published by the Japan Electric Power Exchange (JEPX). Prices are simple averages of 30-minute slots. Our own aggregation.

Balancing market (September 2026, preliminary, nationwide)

Product	Procured volume (MW)	Bid volume (MW)	Awarded volume (MW)	Bid ratio (bids ÷ procured)	Clearing rate (awarded ÷ bids)	Average awarded price (yen/kW per 30 min)	Battery share
Primary reserve (online)	1,388.7	1,710.0	1,337.3	123.1%	78.2%	3.43	27.9%
Primary reserve (offline)	497.8	418.4	265.7	84.0%	63.5%	5.79	100.0%

Product	Procured volume (MW)	Bid volume (MW)	Awarded volume (MW)	Bid ratio (bids ÷ procured)	Clearing rate (awarded ÷ bids)	Average awarded price (yen/kW per 30 min)	Battery share
Secondary reserve 1	962.6	1,996.2	1,102.3	207.4%	55.2%	2.49	1.8%
Secondary reserve 2	983.1	3,709.3	2,163.2	377.3%	58.3%	2.51	1.0%
Tertiary reserve 1	2,498.5	5,895.9	3,021.0	236.0%	51.2%	2.28	0.7%
Tertiary reserve 2	376.3	1,371.7	368.9	364.5%	26.9%	2.22	0.8%
Composite product	3,383.8	6,641.4	3,623.5	196.3%	54.6%	2.64	10.4%

Volumes are averages per 30-minute slot. For products with a bid ratio below 100%, bids fall short of the procured volume. The clearing rate is the share of the bid volume that was awarded.

Source: Trading results (preliminary) and trading results by power source type published by the Electric Power Reserve Exchange (EPRX). Our own aggregation.

Capacity market main auction area prices (yen/kW per year)

Delivery fiscal year	Hokkaido	Tohoku	Tokyo	Chubu	Hokuriku	Kansai	Chugoku	Shikoku	Kyushu
FY2024	14,137	14,137	14,137	14,137	14,137	14,137	14,137	14,137	14,137
FY2025	5,242	3,495	3,495	3,495	3,495	3,495	3,495	3,495	5,242
FY2026	8,749	5,833	5,834	5,832	5,832	5,832	5,832	5,832	8,748
FY2027	13,287	9,044	9,555	7,823	7,638	7,638	7,638	7,638	11,457
FY2028	14,812	14,812	14,812	10,280	8,785	8,785	8,785	8,785	13,177
FY2029	14,972	15,111	15,111	12,388	12,388	12,388	12,388	12,388	15,112

Battery and pumped-hydro capacity contracted in the Long-Term Decarbonization Power Source Auction (10,000 kW)

Auction fiscal year	Category	Contracted capacity (10,000 kW)
FY2023 auction	Batteries and pumped hydro	166.9
FY2024 auction	3–6 hours	96.1
FY2024 auction	6 hours or more	76.9
FY2025 auction	Lithium-ion	81.9
FY2025 auction	Non-lithium	88.6

Source: Clearing results and capacity contract documents published by the Organization for Cross-regional Coordination of Transmission Operators (OCCTO) (sources are listed in the last section (Assumptions and terms)). As published.

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18. Assumptions and terms

- The figures are our own aggregation of the published values from the sources. We have not added up several months or converted anything to an annual basis. We do not show the income of any single battery storage site.
- Months covered: wholesale electricity (JEPX) is October 2026 (31 days); the balancing market (EPRX) is preliminary figures for September 2026 (delivery dates up to the 29th). The balancing market covers a different month until published figures for October 2026 enter our aggregation.
- Average prices are simple averages of 30-minute slots (not weighted by trading volume). Daytime is the slots from 8:00 to 16:00, and evening is the slots from 16:00 to 20:00.
- Spread: we sort the day's 30-minute slots by price and subtract the average of the 8 cheapest slots from the average of the 8 most expensive slots (4 hours) (4 slots for 2 hours). It ignores whether the cheap hours come before the expensive hours, electricity lost in charging and discharging, and fees, taxes and grid usage charges. It does not show actual income.
- Balancing market: nationwide (total column) of EPRX preliminary figures. Procured volume is by grid operator (TSO); bid volume, awarded volume and average awarded price are by the area where the power source is located. Volumes are averages per 30-minute slot. The average awarded price is weighted by awarded volume. Bid ratio = bid volume ÷ procured volume; clearing rate = awarded volume ÷ bid volume. For the 3-hour blocks up to March 13, 2026, we counted 1 block as 6 30-minute slots.
- Range of awarded prices: the thin line runs from the lowest bid (minimum awarded price) to the highest bid (maximum awarded price) among the month's slots with awards (published values by the area where the power source is located). The published values for Primary reserve (online), Secondary reserve 1 and 2, and Tertiary reserve 1 also include Composite product bids (one price for several products together). So we removed from the thin line any bid above 10 yen that can be explained by a Composite product bid in the same slot and area. For Primary reserve (offline), we removed from the thin line bids above the price cap (10 yen this month). (The balancing market system still awards bids above the cap and cuts them down to the cap at settlement.) The thick band is the range from the bottom 10% to 90% when the average awarded prices of slots with awards are sorted from low to high. The dot is the average weighted by awarded volume.
- Clearing rate by grid area × time of day: total awarded volume ÷ total bid volume for slots at the same time of day in the month (by the area where the power source is located).
- The time from instruction to output change and the duration for each balancing market product are the FY2026 specifications. Primary reserve online is the category connected to the grid operator by a dedicated line; offline is the category that runs on the simplified dispatch system without a dedicated line.
- High-price slots for Tertiary reserve 2: slots with a nationwide average awarded price of 10 yen/kW per 30 min or more. The top 1% is the top 1% of slots when ranked by sales (average awarded price × awarded volume). The maximum awarded price is the published value by the area where the power source is located.

Sources (OCCTO, capacity market clearing results): delivery FY2024 https://www.occto.or.jp/iinkai/youryou/kentoukai/2020/files/youryou_kentoukai_27_05.pdf / delivery FY2025 https://www.occto.or.jp/iinkai/youryou/kentoukai/2021/files/youryou_kentoukai_35_03_1.pdf / delivery FY2026 https://www.occto.or.jp/iinkai/youryou/kentoukai/2022/files/youryou_kentoukai_43_03.pdf / delivery FY2027 https://www.occto.or.jp/market-board/market/oshirase/2023/files/240124_mainauction_youryouyakujokekka_kouhyou_jitsujukyu2027.pdf / delivery FY2028 https://www.occto.or.jp/market-board/market/oshirase/2024/files/250129_mainauction_youryouyakujokekka_kouhyou_jitsujukyu2028.pdf / delivery FY2029 https://www.occto.or.jp/assets/news/capacity-market/260120_mainauction_youryouyakujokekka_kouhyou_jitsujukyu2029.pdf

Sources (OCCTO, Long-Term Decarbonization Power Source Auction): FY2023 auction https://www.occto.or.jp/assets/260605_auction_keiyakujyoukyou_kouhyou.pdf / FY2024 auction https://www.occto.or.jp/market-board/market/jitsujukyukanren/files/longauction_keiyakukekka_2024.pdf / FY2025 auction https://www.occto.or.jp/assets/various/capacity-market/jitsujukyukanren/yoryokakuhokeiyaku/longauction_keiyakukekka_2025.pdf

- Capacity market clearing results for each delivery fiscal year were published in different years (FY2024 in 2020, FY2029 in January 2026). We do not show the number of battery projects in the Long-Term Decarbonization Power Source Auction, because it is not in the published summary tables.

Sources: Japan Electric Power Exchange (JEPX) spot market trading results / Electric Power Reserve Exchange (EPRX) trading results (preliminary) and trading results by power source type / Organization for Cross-regional Coordination of Transmission Operators (OCCTO) capacity market main auction clearing results and Long-Term Decarbonization Power Source Auction capacity contract documents / our aggregation of grid connection study answer documents we received / our DealRoom records.

Terms: System price (System, nationwide) = the wholesale electricity price when all of Japan is treated as one market. Area price = a price that splits by area depending on congestion on the power lines. The grid = the power companies' transmission network. Grid connection = connecting to the transmission network. MW = the size of power output (1MW = 1,000kW). MWh = the amount of electricity that can be stored. yen/kWh = the price per 1 kWh of electricity.

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